# Industrial Internship Report on

# File Organizer Using Python

**Prepared by**

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| *Executive Summary* |
|---|
| The Industrial Internship provided by Upskill Campus in collaboration with UniConverge Technologies Pvt. Ltd. (UCT) offered practical exposure to software development using Python. During this internship, I developed a File Organizer application that automatically categorizes files into folders based on their extensions. The project focused on improving file management by reducing manual effort and enhancing organization of digital documents.<br><br>The application scans a selected directory, identifies different file types, creates appropriate folders, and moves files to their respective locations while handling duplicate filenames safely. Throughout the internship I learned project planning, Python programming, debugging, testing, version control using GitHub, and documentation. This project strengthened my understanding of writing clean, modular, and maintainable code while solving a real-world problem. |

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# 1 Preface

Industrial internships bridge the gap between classroom learning and practical software development. This internship provided an opportunity to apply programming concepts to solve a real-world problem.

The assigned project, "File Organizer Using Python," aimed to automate the process of organizing files stored in a directory. Instead of manually sorting documents, images, videos, and other files, the developed application performs the task automatically based on predefined rules.

The internship followed a structured approach beginning with project planning, followed by implementation, testing, documentation, and final submission. Throughout the four-week program I enhanced my Python programming skills, improved my debugging techniques, gained experience with GitHub version control, and learned the importance of writing maintainable code.

I sincerely thank Upskill Campus, UniConverge Technologies Pvt. Ltd., my mentors, and Amity University Jharkhand for providing this valuable learning opportunity. I also thank my family and friends for their continuous encouragement.

I encourage future students to actively participate in industrial internships because they provide practical experience, improve technical confidence, and prepare students for professional software development careers.

# 2 Introduction

## 2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

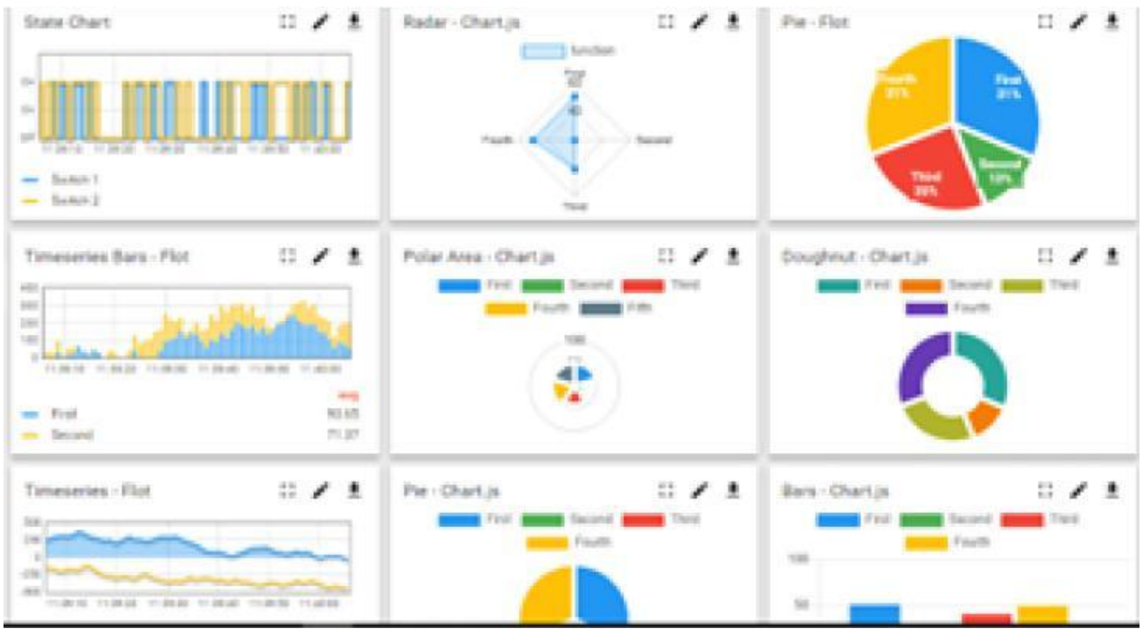

### i. UCT IoT Platform ( uct Insight )

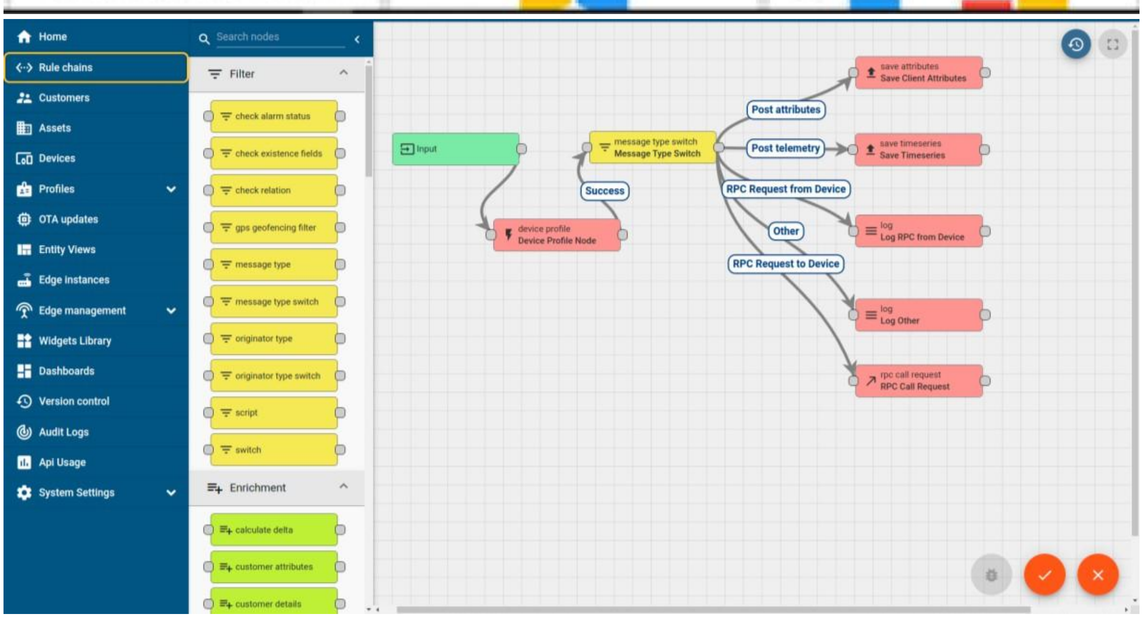
**UCT Insight** is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine





## ii. Smart Factory Platform ( FACTORY WATCH )

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

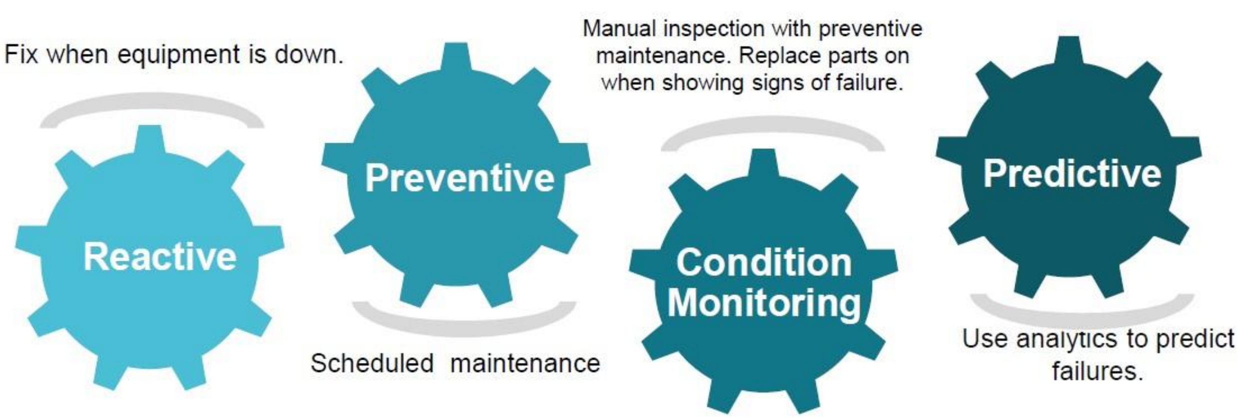
Its unique SaaS model helps users to save time, cost and money.


## iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN ~~teschnology~~technology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

## iv. Predictive Maintenance

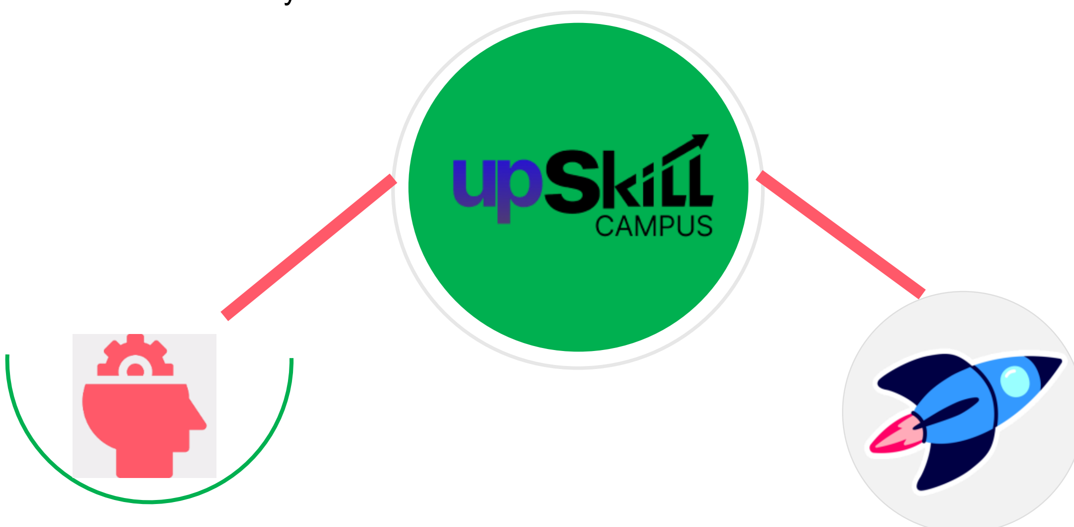
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



## 2.2 About upskill Campus (USC)

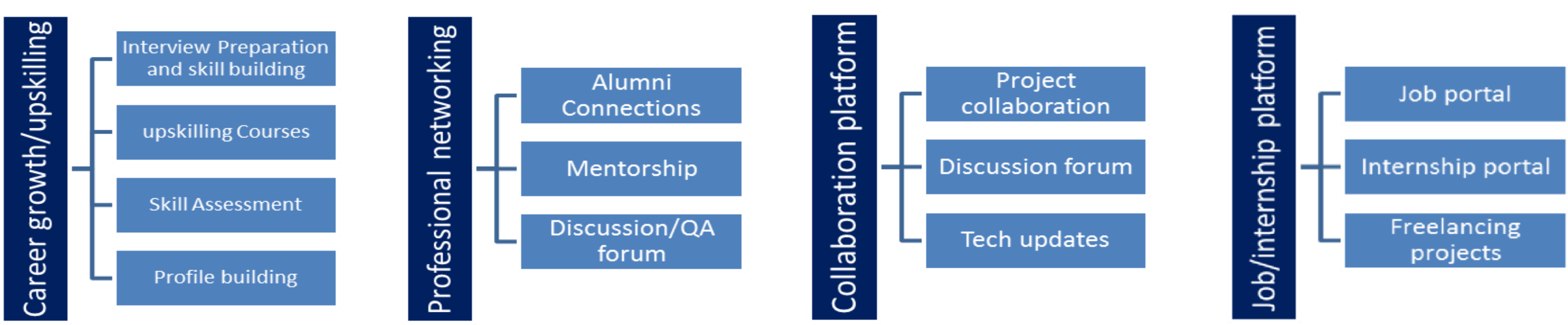
upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.




Seeing need of upskilling in self upSkill Campus aiming paced manner along-with
to upskill 1 million additional support services e.g. learners in next 5 year
Internship, projects, interaction with
Industry experts, Career
growth Services

https://www.upskillcampus.com/



## 2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

## 2.4 Objectives of this Internship program

The objective for this internship program was to ☛

get practical experience of working in the industry.

☛ to solve real world problems.

☛ to have improved job prospects.

☛ to have Improved understanding of our field and its applications.

☛ to have Personal growth like better communication and problem solving.

## 2.5 Reference

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## 2.6 Glossary

Table 1

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## 3 Problem Statement

In the assigned problem statement

Managing files manually becomes difficult when a system contains hundreds of documents, images, videos, compressed files, and applications. Users often waste significant time searching for files because everything is stored in a single folder.

The objective of this project was to design and develop a Python-based File Organizer that automatically scans a selected directory, identifies different file extensions, creates folders according to file categories, and moves files into their appropriate locations. The system also handles duplicate filenames safely and maintains an organized directory structure, improving productivity and reducing manual effort.

# 4 Existing and Proposed solution

Provide summary of existing solutions provided by others, what are their limitations?

**Existing Solution**

Traditional file organization requires users to manually create folders and move files. This process is repetitive, time-consuming, and prone to mistakes. Although operating systems provide basic sorting options, they do not automatically categorize files into separate folders.

**Limitations**

- Time consuming
- Human errors
- Duplicate files
- Poor organization
- Difficult to manage large directories

What is your proposed solution?

**Proposed Solution**

The proposed File Organizer application automates file organization using Python. It scans directories, identifies file extensions, creates folders automatically, and moves files into their appropriate categories while avoiding overwriting duplicate files.

What value addition are you planning?

**Value Addition**

- Automatic organization
- Faster file management
- Reduced human effort
- Safe duplicate handling
- Easy maintenance
- Cross-platform compatibility

### 4.1 Code submission (Github link)

https://github.com/Amartya1103/Internship-program-by-upSkillcampus/blob/main/File%20Organizer.py

4.2 **Report submission (Github link):**

first make placeholder, copy the link.

https://github.com/Amartya1103/Internship-program-by-upSkillcampus/blob/main/main.py

# 5 Proposed Design/ Model

Given more details about design flow of your solution. This is applicable for all domains. DS/ML Students can cover it after they have their algorithm implementation. There is always a start, intermediate stages and then final outcome.

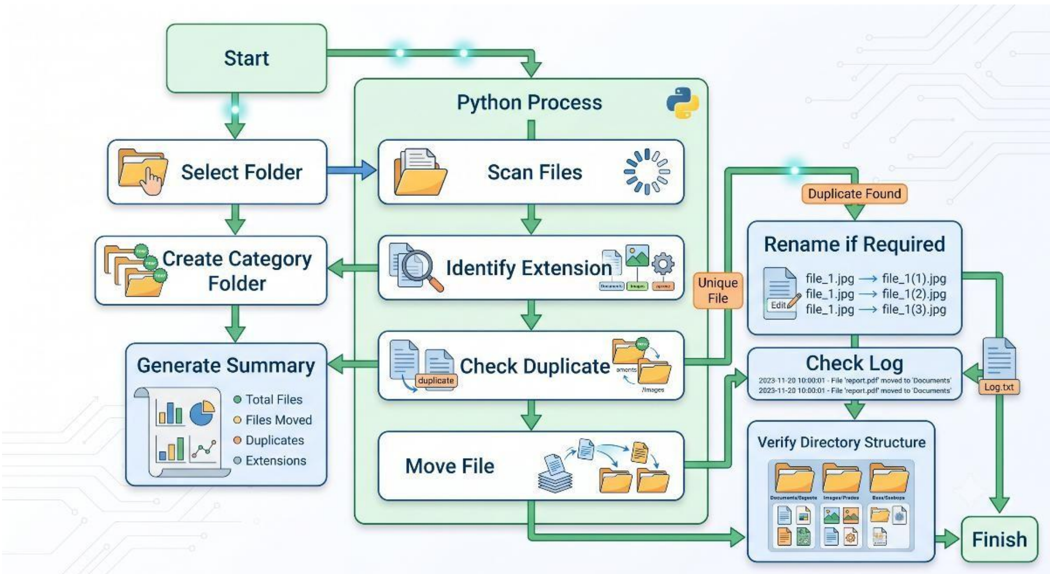


**Figure 1: WORK FLOW DIAGRAM OF THE SYSTEM**

## 5.1 High Level Diagram (if applicable)

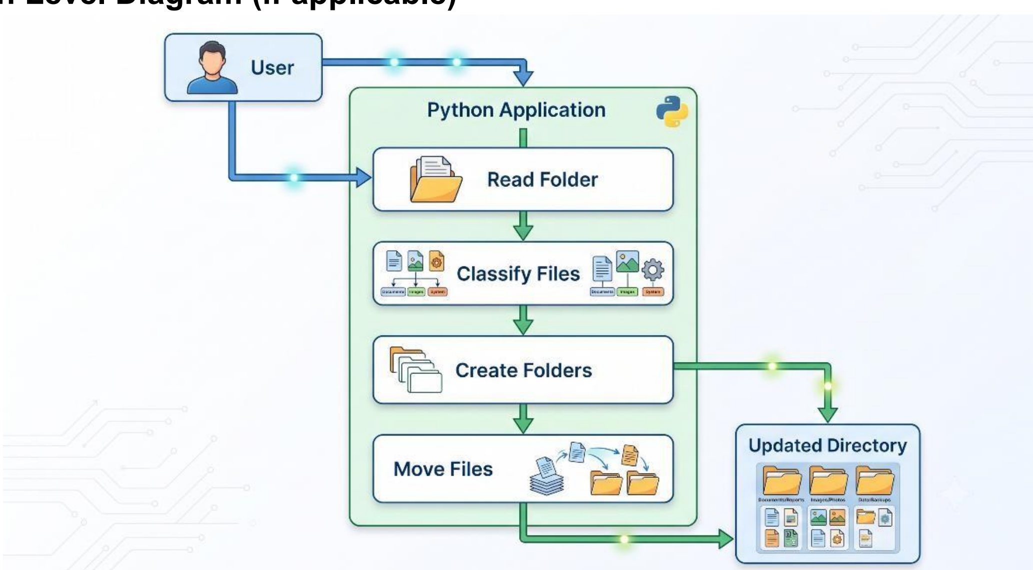


**Figure 2: HIGH LEVEL DIAGRAM OF THE SYSTEM**

## 5.2 Low Level Diagram (if applicable)

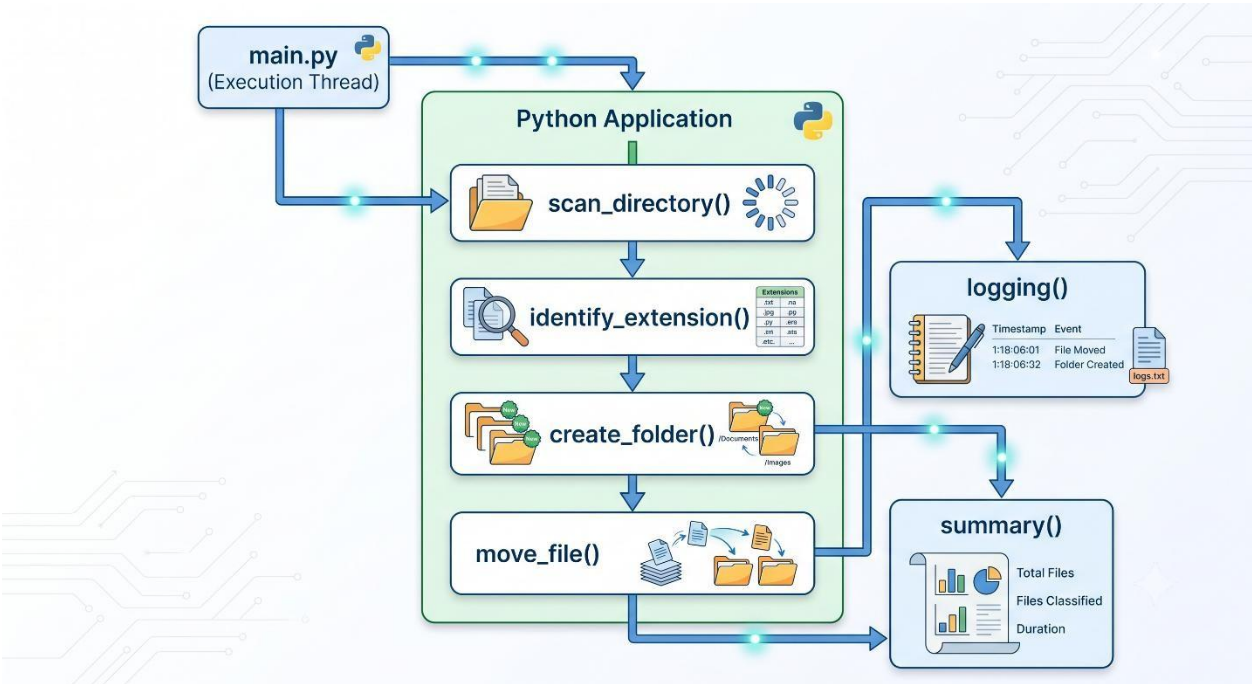


**Figure 3: LOW LEVEL DIAGRAM OF THE SYSTEM**

## 5.3 Interfaces (if applicable)

Update with Block Diagrams, Data flow, protocols, FLOW Charts, State Machines, Memory Buffer Management.

# 6 Performance Test

This is very important part and defines why this work is meant of Real industries, instead of being just academic project.

Here we need to first find the constraints.

How those constraints were taken care in your design?

What were test results around those constraints?

Constraints can be e.g. memory, MIPS (speed, operations per second), accuracy, durability, power consumption etc.

In case you could not test them, but still you should mention how identified constraints can impact your design, and what are recommendations to handle them.

## 6.1 Test Plan/ Test Cases

Table 2

| Test Case | Expected Result |
|---|---|
| Empty Folder | No error |
| Mixed Files | Files categorized |
| Duplicate File | Renamed safely |
| Unknown Extension | Other folder |
| Large Folder | Organized successfully |

## 6.2 Test Procedure

The application was tested using directories containing different file formats such as PDF, DOCX, JPG, PNG, MP4, ZIP, and Python files. Duplicate filenames and empty folders were also tested to verify application stability.

## 6.3 Performance Outcome

The application successfully categorized files based on extensions, created folders automatically, handled duplicate filenames without overwriting existing files, and completed execution within a few seconds for directories containing hundreds of files.

# 7 My learnings

You should provide summary of your overall learning and how it would help you in your career growth.

This internship enhanced my practical understanding of Python programming and software development. I learned how to structure projects, use Python libraries for file management, implement error handling, test applications, document projects, and maintain code using GitHub. The experience improved my analytical thinking, debugging skills, and confidence in developing real-world software solutions. These skills will be valuable for future internships and software engineering roles.

# 8 Future work scope

You can put some ideas that you could not work due to time limitation but can be taken in future.

Future improvements for the File Organizer include:

- AI-based file categorization
- Duplicate file detection using hash algorithms
- Automatic scheduled organization
- Cloud storage integration
- File search functionality
- File encryption
- Desktop GUI enhancements
- Activity logs and analytics dashboard